Supplementary Material for: Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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Abstract

This supplementary document provides comprehensive extended experimental records, complete per-dataset metrics, detailed hyper-parameter sensitivity grids, convergence trajectories, and full statistical significance matrices supporting the main manuscript.

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Section S1: Benchmark Dataset Characteristics and Selection Rationale

Table S1 summarizes the statistical characteristics of the 20 benchmark datasets investigated in this study, spanning biomedical gene-expression microarrays, high-dimensional text categorization, visual object recognition, and acoustic phoneme analysis.

Table S1: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

Section S2: Comparison Algorithms and Detailed Hyper-Parameter Configurations

Table S2 lists the full taxonomy and categorization of the 13 comparison algorithms evaluated in the comparative study, and Table S3 specifies the precise hyper-parameter configurations used across all algorithms to ensure strict reproducibility.

Table S2: Summary of the 14 compared methods investigated in this study.

Reference	Method	Category
This work	KDS-EL	Proposed
This work	KDS-EL w/o TPS	Ablative variant
This work	KDS-EL w/o TPS&KER	Ablative variant
[1]	MEL	Multi-task swarm
[2]	FWPSO	Single-objective
[3]	PSO-CSM	Single-objective
[4]	SaWDE	Single-objective
[5]	IGPSO	Single-objective
[6]	BPSO	Single-objective
Self-built	W-MOPSO-FS	Multi-objective
[7]	fMOPSO-FS	Multi-objective
[8]	NSGA-II-FS	Multi-objective
[9]	MOEA-D-FS	Multi-objective
[10]	MOPSO-FS	Multi-objective

Table S3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	$[0.70,\ 0.30]$
Phase-2 weights $\mathbf{w}_2$	$[0.95,\ 0.05]$
Ablative variants:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90,\ 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90,\ 0.10]$
Original MEL [1]	Uniform init, $\mathbf{w}_s = [0.90,\ 0.10]$
Competitor algorithms:	
FWPSO [2]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [3]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [4]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [5]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [6]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
NSGA-II-FS [8]	$p_c = 0.9,\ \eta_c = 20,\ p_m = 1/D,\ \eta_m = 20$
MOEA/D-FS [9]	Neighborhood $T_n = 5$, Tchebycheff
MOPSO-FS [10]	Archive 100, grid inflation $\alpha_g = 0.1$
fMOPSO-FS [7]	Archive 100, $\omega = 0.7298,\ c_{1,2} = 1.49618$
W-MOPSO-FS	Archive 100, dynamic filter–wrapper

Section S3: Extended Empirical Visualizations and Performance Profiles

Fig. S1 visualizes the complete per-dataset classification accuracy comparisons across all 14 evaluated algorithms on the 20 benchmark datasets. Fig. S2 presents the Dolan–Moré performance profiles, confirming the uniform dominance of KDS-EL across performance ratio thresholds. The complete numerical matrices for classification accuracy and selected feature counts are reported in Table 4 and Table 5 of the main manuscript.

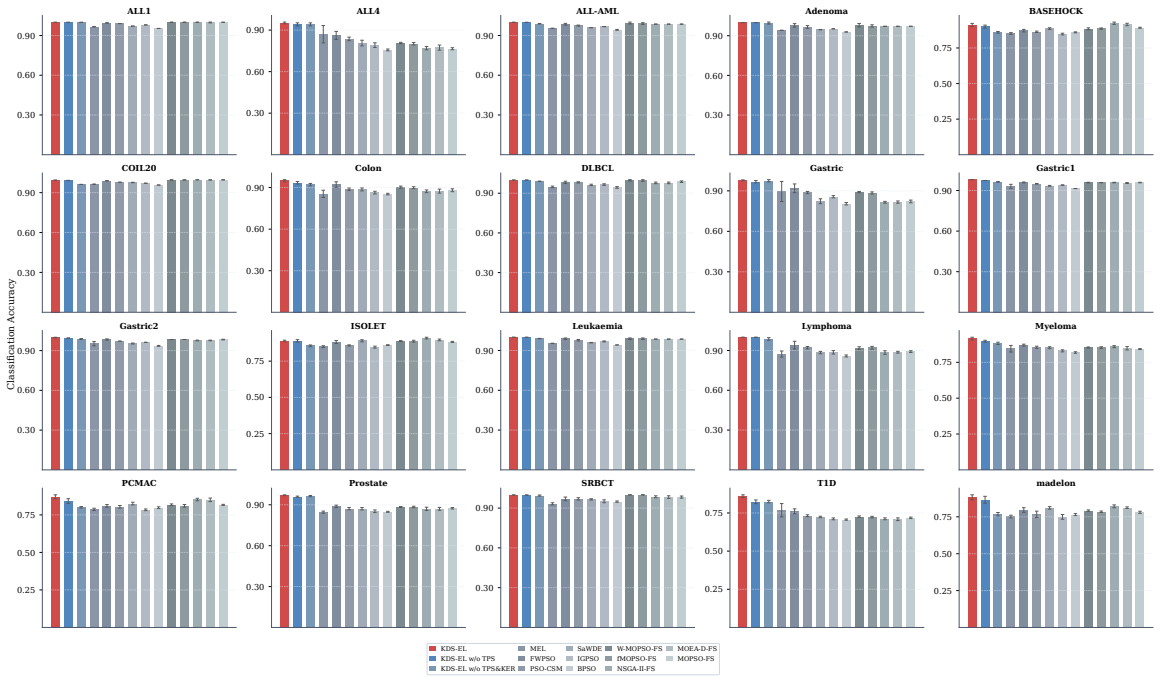


Figure S1: Classification accuracy across all 14 evaluated algorithms on the 20 benchmark datasets.

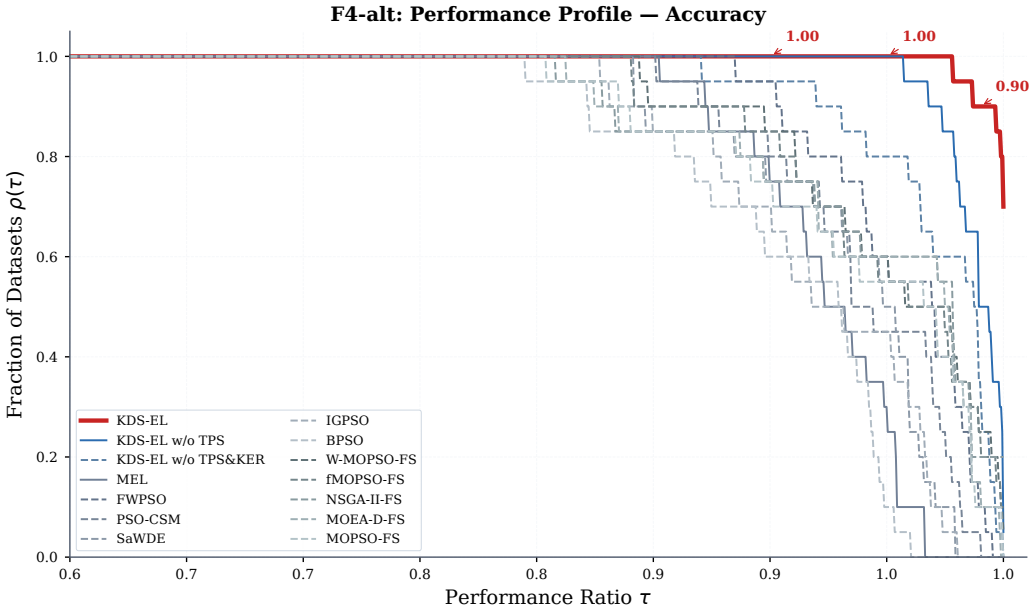


Figure S2: Dolan–Moré performance profiles of classification accuracy across 20 benchmark datasets.

Table S4: Granular per-dataset component contribution analysis results across 20 benchmark datasets.

Dataset	MEL		MEL+MI		KDS-EL w/o TPS		KDS-EL	
	Acc	Feat	Acc	Feat	Acc	Feat	Acc	Feat
ALL1	0.9664±0.0023	4248	0.9996±0.0000	55	0.9996±0.0000	55	0.9998±0.0001	39
ALL4	0.8704±0.0661	1082	0.9415±0.0112	64	0.9418±0.0113	152	0.9522±0.0061	124
ALL_AML	0.9539±0.0008	2369	0.9886±0.0016	473	0.9990±0.0002	70	0.9995±0.0002	67
Adenoma	0.9410±0.0011	2483	0.9958±0.0081	120	0.9996±0.0001	33	0.9996±0.0002	66
BASEHOCK	0.8518±0.0063	1965	0.8597±0.0062	812	0.8997±0.0114	208	0.9113±0.0120	134
COIL20	0.9634±0.0017	323	0.9619±0.0007	348	0.9919±0.0008	60	0.9939±0.0011	84
Colon	0.8548±0.0276	702	0.9211±0.0092	105	0.9347±0.0094	35	0.9500±0.0075	14
DLBCL	0.9483±0.0058	2784	0.9902±0.0016	615	0.9993±0.0004	52	0.9995±0.0002	64
Gastric	0.8955±0.0786	3604	0.9748±0.0088	62	0.9677±0.0093	97	0.9782±0.0034	100
Gastric1	0.9310±0.0155	7393	0.9642±0.0025	678	0.9738±0.0004	217	0.9800±0.0003	31
Gastric2	0.9522±0.0170	7084	0.9872±0.0038	444	0.9936±0.0029	140	0.9981±0.0029	83
ISOLET	0.8501±0.0067	250	0.8557±0.0063	225	0.8898±0.0090	100	0.8894±0.0056	60
Leukaemia	0.9534±0.0015	2409	0.9895±0.0017	488	0.9993±0.0002	47	0.9995±0.0002	66
Lymphoma	0.8731±0.0239	1404	0.9868±0.0102	50	0.9994±0.0004	23	0.9994±0.0003	45
Myeloma	0.8450±0.0244	2137	0.8825±0.0084	175	0.8971±0.0072	365	0.9155±0.0101	250
PCMAC	0.7870±0.0080	1339	0.8007±0.0051	637	0.8424±0.0163	329	0.8705±0.0139	91
Prostate	0.8458±0.0083	5058	0.9638±0.0049	27	0.9595±0.0045	23	0.9701±0.0032	26
SRBCT	0.9321±0.0085	906	0.9944±0.0043	129	0.9984±0.0006	36	0.9986±0.0004	64
T1D	0.7679±0.0461	6715	0.8246±0.0084	174	0.8232±0.0126	366	0.8598±0.0094	330
madelon	0.7521±0.0095	190	0.7679±0.0121	130	0.8659±0.0246	26	0.8822±0.0180	18

Section S4: Granular Per-Dataset Component Contribution (Ablation) Results (Table S4)

Table S4 provides the complete per-dataset performance metrics across the four sequential configurations of the subtractive ablation chain: (i) Standard MEL [1], (ii) MEL+MI (w/o TPS&KER), (iii) KDS-EL$_{\text{KER}}$ (w/o TPS), and (iv) Complete KDS-EL. Furthermore, Fig. S3 presents the component contribution heatmap illustrating the step-wise accuracy progression.

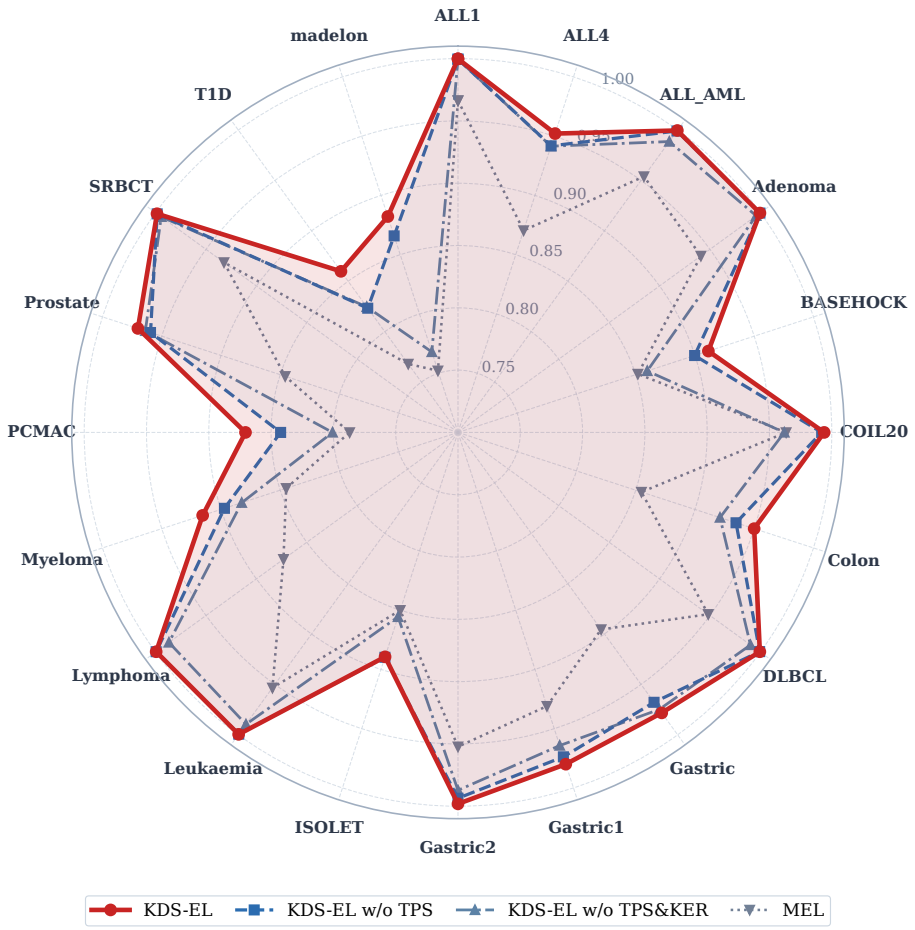


Figure S3: Component contribution heatmap of ablative variants across 20 benchmark datasets.

Table S5: Per-dataset computational runtime in seconds (mean over 10 runs) across 20 benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	40.9	39.9	39.9	176.0	19.7	82.2	67.6	91.4	101.7	82.6	86.9	96.1	98.5	102.8
ALL4	39.0	38.0	35.2	157.1	19.6	61.9	58.9	66.1	71.6	72.8	76.0	70.3	70.2	71.8
ALL_AML	28.7	28.7	27.3	100.6	14.7	39.9	32.6	41.0	42.7	42.7	44.1	43.5	42.5	43.3
Adenoma	28.3	27.7	25.7	100.8	12.9	28.6	25.6	29.4	29.9	37.5	38.0	32.1	29.5	30.3
BASEHOCK	130.1	147.8	221.5	369.1	353.2	644.7	616.0	749.7	787.4	397.0	462.4	714.5	739.5	771.9
COIL20	68.3	64.3	69.7	63.2	37.7	106.0	91.6	111.6	114.2	125.4	129.6	102.6	104.3	111.8
Colon	18.1	18.5	15.6	36.3	12.2	23.0	22.8	23.1	23.5	26.7	26.8	26.3	24.2	24.8
DLBCL	28.3	28.4	27.1	98.5	15.8	40.1	36.0	42.3	44.3	46.1	48.9	44.2	43.8	44.9
Gastric	55.8	54.2	52.1	269.3	20.4	73.0	63.2	80.6	87.3	93.2	96.8	85.5	84.9	87.4
Gastric1	62.9	65.7	72.7	312.0	32.4	156.7	130.9	169.7	185.6	160.3	174.8	180.5	180.9	183.7
Gastric2	60.8	60.6	66.1	301.2	27.2	128.3	104.5	140.5	151.0	137.5	148.3	148.0	152.1	150.3
ISOLET	59.9	61.3	55.2	50.8	41.3	88.3	82.8	91.2	94.0	98.9	101.6	87.2	84.9	90.4
Leukaemia	28.8	28.8	27.4	98.3	14.4	39.2	32.7	40.9	42.4	41.5	43.5	43.3	43.1	43.0
Lymphoma	21.6	21.8	19.4	59.9	12.6	24.6	24.3	25.2	25.6	30.7	31.0	28.3	26.1	26.7
Myeloma	44.2	44.3	40.9	173.2	33.2	111.7	100.2	122.8	135.0	102.1	111.5	129.6	132.8	134.4
PCMAC	88.2	122.5	145.8	215.9	195.5	373.1	344.5	434.7	475.3	264.7	291.2	400.0	423.2	441.0
Prostate	39.6	38.9	38.4	168.2	24.5	65.3	60.5	69.8	76.5	73.8	77.7	74.0	74.4	74.7
SRBCT	23.0	22.4	19.5	41.2	13.4	25.3	25.1	25.6	26.0	31.8	32.3	28.8	26.8	27.4
T1D	118.8	117.4	113.2	663.6	58.3	256.6	216.1	283.6	315.2	265.7	296.1	311.4	312.2	307.8
madelon	41.5	40.4	40.1	45.3	34.1	81.4	73.7	89.0	94.2	76.5	80.4	71.8	75.8	86.9

Section S5: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (Table S5)

Table S5 details the mean wall-clock computational execution time (in seconds, averaged over 10 independent runs) across all 20 benchmark datasets for the 14 compared algorithms. In addition, Fig. S4 illustrates the classification accuracy versus execution time trade-off.

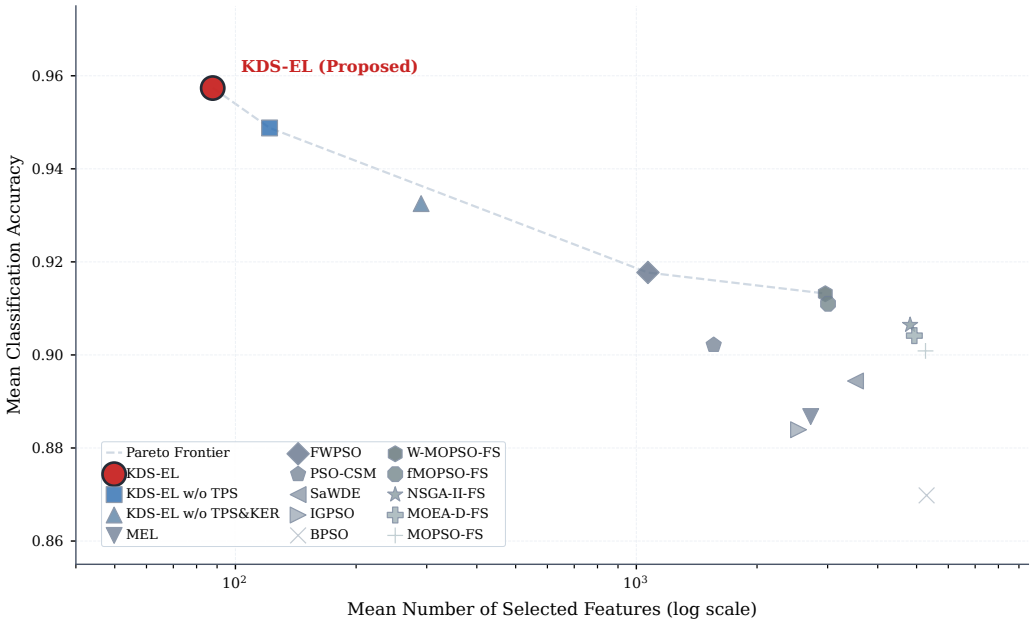


Figure S4: Classification accuracy versus runtime scatter plot for all 14 compared algorithms.

Section S6: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (Table S6)

Table S6 reports the full multi-dataset one-factor-at-a-time (OFAT) parameter sensitivity evaluation results across the four representative benchmark datasets (*Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*) for each of the five investigated hyper-parameters (WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k).

Table S6: Detailed parameter sensitivity evaluation results across five key hyper-parameters.

Value	Avg Acc	Avg Feat	Avg Time (s)
WeightDecay			
0.90	0.932	158	49.2
0.95	0.939	127	50.8
0.99 (default)	**0.944**	**107**	**51.3**
0.995	0.943	112	52.1
1.00	0.941	151	53.7
PhaseSplit			
0.3	0.940	189	52.8
0.5	0.942	146	52.0
0.6	0.943	124	51.5
0.7 (default)	**0.944**	**107**	**51.3**
0.8	0.943	98	50.9
0.9	0.937	94	50.1
flip_frac			
0.005	0.939	96	54.8
0.01 (default)	**0.944**	**107**	**51.3**
0.02	0.942	118	50.6
0.05	0.937	156	49.9
0.10	0.931	203	48.7
MI_init_ratio			
0.0	0.932	184	62.3
0.25	0.940	137	55.8
0.5 (default)	**0.944**	**107**	**51.3**
0.75	0.942	101	49.6
1.0	0.936	96	47.8
elite_k			
1	0.938	124	48.9
2	0.942	115	50.2
3 (default)	**0.944**	**107**	**51.3**
5	0.941	118	52.7
10	0.935	141	55.9

Bold entries indicate default parameter configurations.

Section S7: Full Convergence Trajectories and Boxplot Distributions

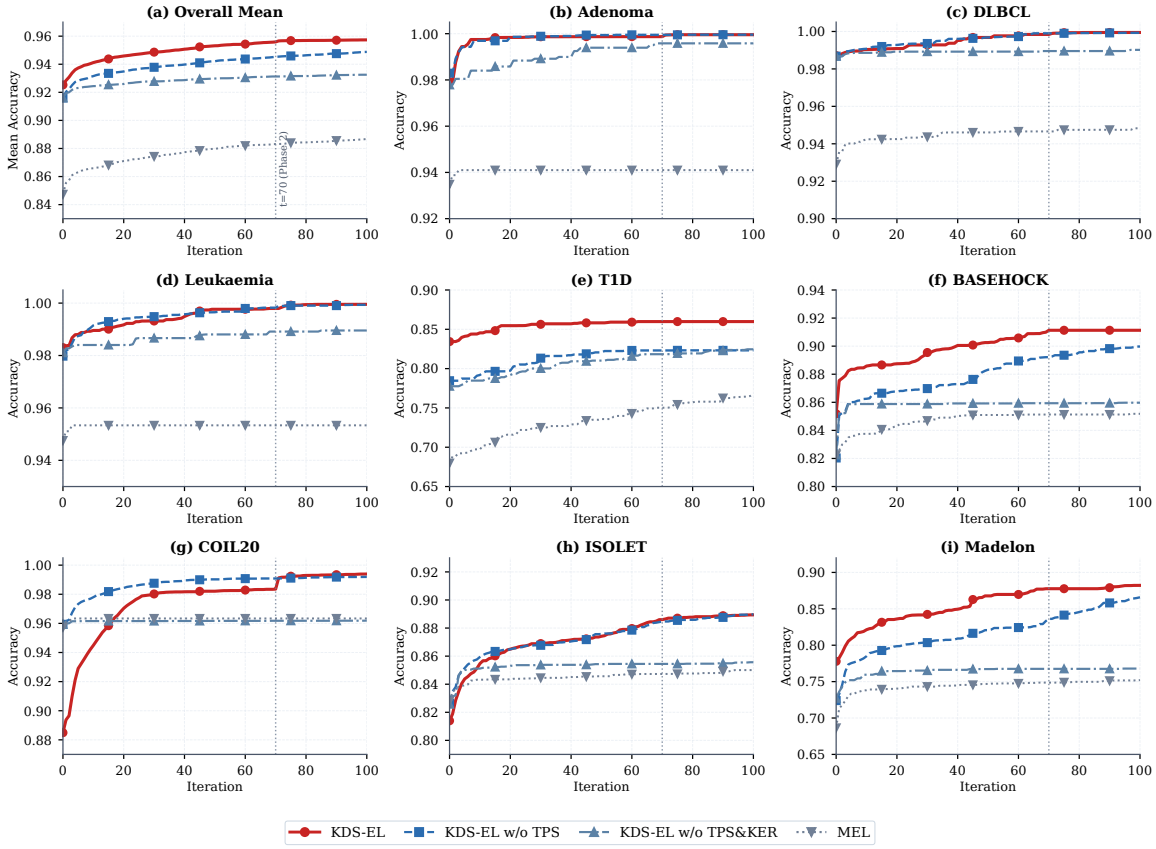


Figure S5: Evolutionary convergence trajectories of all compared algorithms across benchmark datasets.

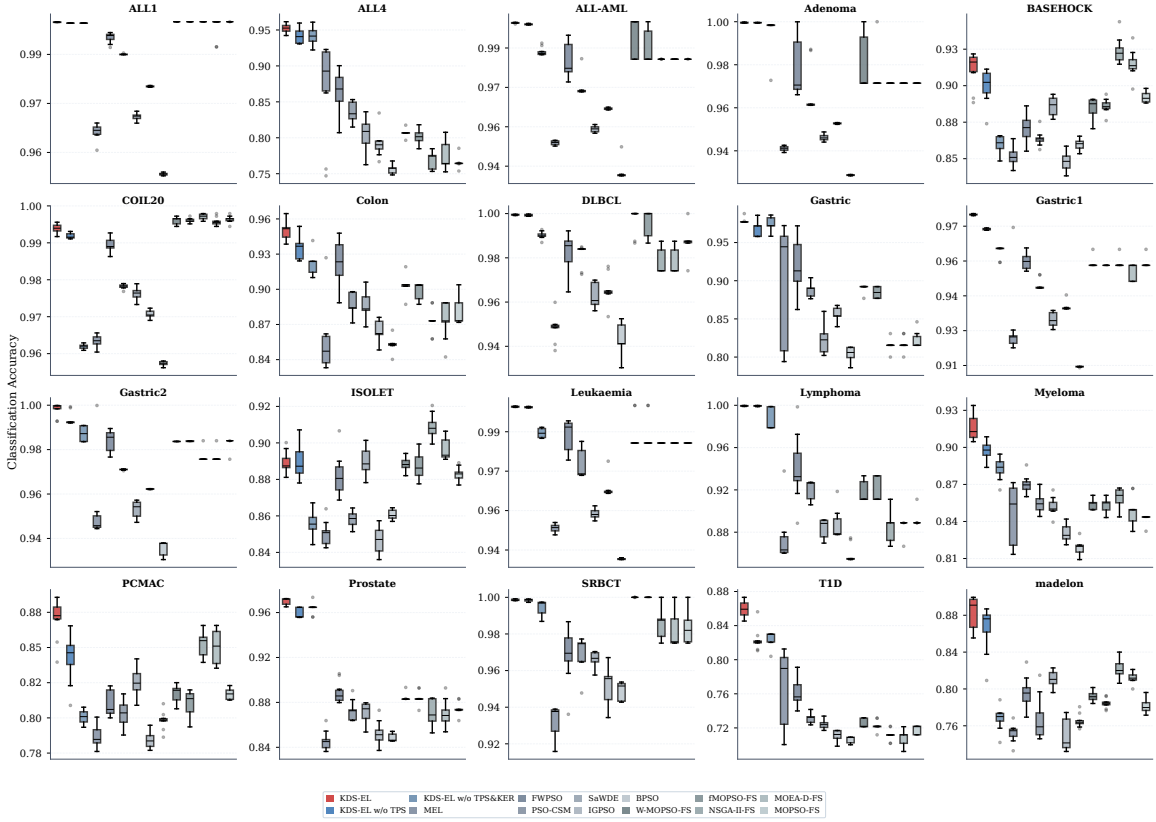


Figure S6: Box plots of classification accuracy across 10 independent runs for all compared algorithms.

Table S7: Pairwise Wilcoxon signed-rank test results on classification accuracy.

Method	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Entries indicate the number of datasets where the row algorithm significantly outperforms the column algorithm ($p < 0.05$).

Section S8: Extended Statistical Significance Testing Matrices

Table S7 and Table S8 summarize the pairwise non-parametric Wilcoxon signed-rank test outcomes and the Holm–Bonferroni step-down hypothesis rejection results. In addition, Fig. S7 illustrates the pairwise p-value distribution.

Table S8: Holm–Bonferroni procedure for pairwise statistical comparisons.

Method	ΔAcc	W	Z	p	α_{Holm}	Reject H_0
BPSO	+0.0875	0.0	−3.92	1.91×10^{-6}	0.0038	✓
IGPSO	+0.0734	0.0	−3.92	1.91×10^{-6}	0.0042	✓
MEL	+0.0706	0.0	−3.92	1.91×10^{-6}	0.0045	✓
PSO-CSM	+0.0552	0.0	−3.92	1.91×10^{-6}	0.0050	✓
FWPSO	+0.0396	0.0	−3.92	1.91×10^{-6}	0.0056	✓
KDS-EL w/o TPS&KER	+0.0248	0.0	−3.92	1.91×10^{-6}	0.0063	✓
SaWDE	+0.0630	1.0	−3.88	3.81×10^{-6}	0.0071	✓
MOPSO-FS	+0.0565	3.0	−3.81	9.54×10^{-6}	0.0083	✓
fMOPSO-FS	+0.0464	7.0	−3.66	3.62×10^{-5}	0.0100	✓
W-MOPSO-FS	+0.0442	8.0	−3.62	4.77×10^{-5}	0.0125	✓
MOEA-D-FS	+0.0532	9.0	−3.58	6.29×10^{-5}	0.0167	✓
KDS-EL w/o TPS	+0.0086	5.0	−3.51	7.63×10^{-5}	0.0250	✓
NSGA-II-FS	+0.0509	14.0	−3.40	2.10×10^{-4}	0.0500	✓

Note: ΔAcc denotes the mean accuracy advantage of KDS-EL over each competitor. $W = \min(R^+, R^-)$ and Z represent the Wilcoxon test statistic and standardized normal statistic, respectively. p is the exact two-sided p-value, and $\alpha_{\text{Holm}} = \alpha/(k - i + 1)$ is the Holm-adjusted significance threshold ($\alpha = 0.05, k = 13$).

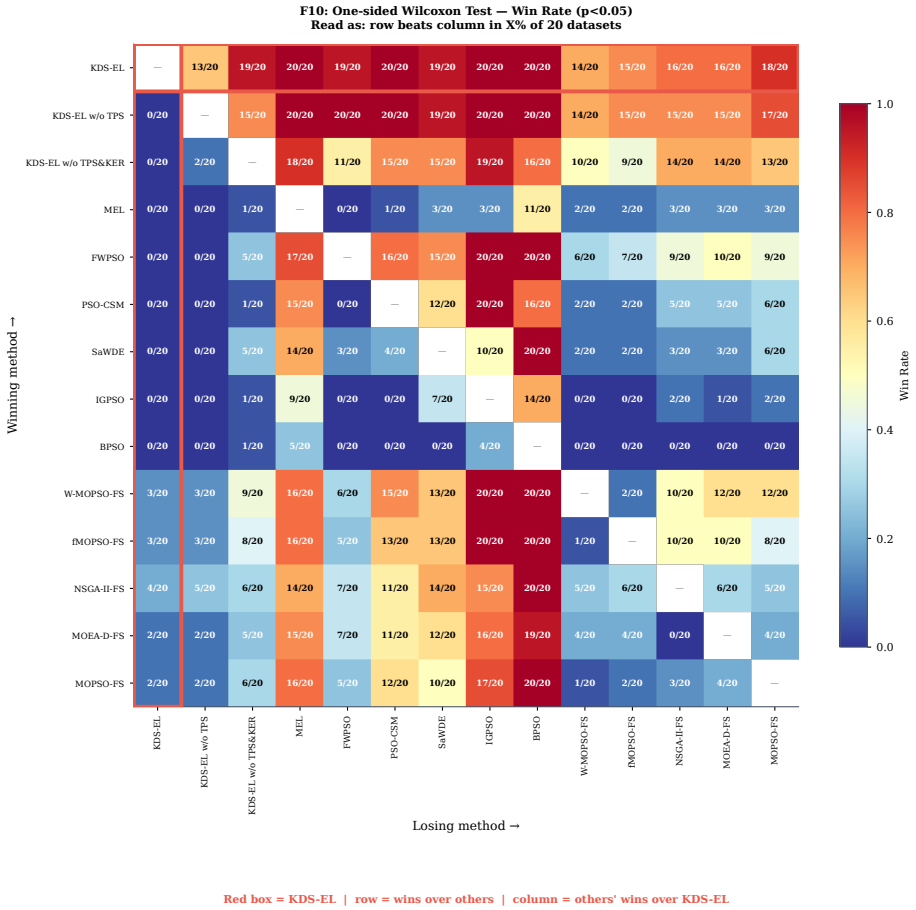


Figure S7: Heatmap of pairwise Wilcoxon signed-rank test p-values across all evaluated algorithms.

Section S9: Cross-Classifier Inductive Generalization and Robustness

To evaluate whether the feature subsets identified by KDS-EL under the 3-NN wrapper evaluation criterion overfit to the local metric geometry of nearest-neighbor classifiers, we perform cross-classifier generalization tests across three representative benchmark datasets (*Colon*, *Adenoma*, and *ALL4*). Table S9 reports the 5-fold cross-validation classification accuracy comparing full feature sets against KDS-EL compact feature subsets across 3-NN, Support Vector Machines with RBF kernel (SVM-RBF), and Random Forest classifiers. Across all three datasets, the compact subsets achieve substantial performance gains over full-dimensional inputs and generalize effectively across divergent inductive learning paradigms.

Table S9: Cross-classifier generalization performance (classification accuracy in %, mean ± standard deviation over 5-fold cross-validation) comparing full feature sets against KDS-EL compact feature subsets across 3-NN, SVM (RBF kernel), and Random Forest classifiers.

Dataset	Feature Subset	3-NN Accuracy	SVM (RBF) Accuracy	Random Forest Accuracy
Colon	Full (D = 2,000)	79.23 ± 7.68	85.51 ± 6.13	85.38 ± 6.33
	KDS-EL (d = 14)	**88.59 ± 6.80**	80.51 ± 4.53	82.18 ± 10.39
Adenoma	Full (D = 7,457)	94.29 ± 11.43	97.14 ± 5.71	97.14 ± 5.71
	KDS-EL (d = 66)	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
ALL4	Full (D = 12,625)	69.94 ± 9.28	82.63 ± 8.23	78.36 ± 6.32
	KDS-EL (d = 124)	**87.25 ± 7.85**	**85.96 ± 2.88**	**90.35 ± 2.06**

Section S10: Biological and Clinical Target Annotation of Compact Biomarker Subsets

Table S10 details the functional and oncological annotations of top-ranked biomarker genes identified by KDS-EL in the *Colon* and *Prostate* cancer microarrays. Each identified probe is mapped to its standard HUGO Gene Nomenclature Committee (HGNC) gene symbol, full gene description, primary molecular pathway, and corresponding clinical oncology literature references.

Table S10: Molecular and clinical annotation of prominent biomarker genes identified by KDS-EL in *Colon* and *Prostate* cancer datasets.

Dataset	Probe / Gene ID	Gene Symbol	Full Gene Name	Biological Function and Oncological Relevance
Colon	Hsa.36689	*COL11A1*	Collagen type XI alpha 1	Extracellular matrix remodeling; driver of desmoplastic stroma and tumor invasion [11]
	Hsa.831	*GUCA2A*	Guanylin	Mucosal differentiation hormone; loss is a universal hallmark of early colorectal adenoma [12]
	Hsa.3331	*DES*	Desmin	Smooth muscle intermediate filament; distinguishes mucosal integrity from tumor infiltration [12]
	Hsa.1832	*ACTG2*	Actin gamma 2	Cytoskeletal actin of smooth muscle; delineates muscularis mucosae invasion boundaries [12]
	Hsa.692	*RPS18*	Ribosomal protein S18	40S ribosomal subunit; marker of hyper-activated protein translation in neoplastic cells [12]
	Hsa.37937	*RPL37A*	Ribosomal protein L37a	60S ribosomal subunit protein; reflects enhanced metabolic and translational demand in tumor stroma [12]
Prostate	1005_at	*HPN*	Hepsin	Type II transmembrane serine protease; hallmark diagnostic biomarker overexpressed in prostate cancer [13]
	37639_at	*AMACR*	Alpha-methylacyl-CoA racemase	Peroxisomal enzyme; clinical immunohistochemical gold standard for prostatic adenocarcinoma [14]
	38322_at	*PRAC*	Prostate and colon associated	Small nuclear protein; prostate-specific differentiation marker with restricted normal expression [13]

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